

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     |  |  |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--------------------|
| 24UAM512                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MACHINE LEARNING                                                                                                                                    |  |  | L/T/P/C<br>3/0/2/4 |
| Nature of Course: Laboratory Integrated Theory                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                     |  |  | PCC                |
| Prerequisite(s): Basics of Python Programming, Statistics and Numerical Methods, Algebra and Combinatorics and Foundations of Data Science.                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                     |  |  |                    |
| Course Objectives: Enable the students to                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                     |  |  |                    |
| ➤                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Apply fundamental linear algebra and probability concepts to mathematically represent data and perform probabilistic reasoning in machine learning. |  |  |                    |
| ➤                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Apply supervised learning algorithms and evaluate models using key classification metrics.                                                          |  |  |                    |
| ➤                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Apply key unsupervised learning techniques, probabilistic models, and Bayesian inference methods for data analysis and pattern discovery.           |  |  |                    |
| ➤                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Design and optimize neural network models using propagation, activation, loss, tuning, and regularization techniques for real-world applications.   |  |  |                    |
| ➤                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Analyze the computational complexity and scalability of machine learning algorithms and optimization methods.                                       |  |  |                    |
| Course Contents:                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                     |  |  |                    |
| UNIT I                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MATHEMATICAL FOUNDATIONS OF MACHINE LEARNING                                                                                                        |  |  | CO19               |
| Linear Algebra-Vector and matrix operations, eigenvalues/eigenvectors, and Singular Value Decomposition (SVD). Probability & Statistics - Random variables, key distributions, expectation, variance, conditional probability, and Bayesian inference. Optimization & Learning Theory- Convexity, loss functions, gradient descent, hypothesis spaces, inductive bias, generalization, overfitting, and bias-variance trade-off.                                 |                                                                                                                                                     |  |  |                    |
| UNIT II                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SUPERVISED LEARNING                                                                                                                                 |  |  | CO29               |
| Linear Regression with One and Multiple Variables-Logistic Regression for Classification- k-Nearest Neighbors (k-NN) - Naïve Bayes Classifier-Support Vector Machines (SVM)-Decision Trees & Random Forests. Evaluation Metrics- Confusion Matrix, Precision, Recall, and F1-Score, ROC Curve and AUC                                                                                                                                                            |                                                                                                                                                     |  |  |                    |
| UNIT III                                                                                                                                                                                                                                                                                                                                                                                                                                                         | UNSUPERVISED LEARNING                                                                                                                               |  |  | CO39               |
| k-Means Clustering-Hierarchical Clustering-Density based clustering- Gaussian Mixture Models (GMM), Principal Component Analysis (PCA)- Bayesian Networks-.Hidden Markov Models (HMM)-Bayesian Inference Methods.                                                                                                                                                                                                                                                |                                                                                                                                                     |  |  |                    |
| UNIT IV                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NEURAL NETWORK                                                                                                                                      |  |  | CO 49              |
| Structure of a Neural Network-Activation Functions-Forward propagation-Loss Functions-Back propagation-Hyper parameter tuning-Over fitting and under fitting. Introduction to Regularization, Applications: Digit recognition (MNIST) and Binary classification tasks.                                                                                                                                                                                           |                                                                                                                                                     |  |  |                    |
| UNIT V                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MODEL OPTIMIZATION, ENSEMBLE METHODS, AND APPLICATIONS                                                                                              |  |  | CO 59              |
| Cross-validation: k-fold, stratified , Parameter Tuning: Grid search, random search , Metrics & Error Correction Bagging and Boosting: AdaBoost, XGBoost , Basics of Reinforcement Learning & RL Framework , Markov Decision Process,Exploration vs.Exploitation Policies,Value Functions and Bellman Equations,Applications-Personalized Medicine and Treatment Optimization,Advanced Conversational Agents (Chatbots, Virtual Assistants), Autonomous Vehicles |                                                                                                                                                     |  |  |                    |
| Theory Hours:                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |  |  | 45                 |

| PRACTICAL EXERCISES |                                                                                                                                                    |           |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1.                  | Apply SVD to a grayscale image and reconstruct the image using 10, 30, and 50 singular values. Compare image quality and storage reduction.        | CO1       |
| 2.                  | Implement Linear Regression using Scikit-learn and NumPy to predict house prices based on area and number of rooms using a sample housing dataset. | CO2       |
| 3.                  | Apply Linear, Polynomial, and RBF SVM kernels using Scikit-learn to classify Iris flower species and compare their classification accuracy.        | CO2       |
| 4.                  | Implement Decision Trees and Random Forests on a classification problem                                                                            | CO2       |
| 5.                  | Implement k-Means clustering on an unlabeled dataset and visualize cluster formation.                                                              | CO3       |
| 6.                  | Analyze and compare the training time, inference time, and prediction performance of different machine learning algorithms using Housing Dataset   | CO4       |
| 7.                  | Analyze the effectiveness of boosting algorithms for heart disease prediction using the Heart Disease Dataset.                                     | CO5       |
| 8.                  | Open Ended Experiment                                                                                                                              | (CO1-CO5) |
| Practical Hours:    |                                                                                                                                                    | 30        |
| Total Hours:        |                                                                                                                                                    | 75        |

| COURSE OUTCOMES                                   |                                                                                                                                                                                      |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At the end of the course students will be able to |                                                                                                                                                                                      |
| CO1                                               | Apply concepts of linear algebra, probability, statistics, and optimization techniques to mathematically represent data and analyze the foundations of machine learning algorithms.  |
| CO2                                               | Apply key supervised learning models and evaluate their performance using essential Classification metrics for informed decision-making.                                             |
| CO3                                               | Apply unsupervised learning algorithms, probabilistic graphical models, and perform Bayesian inference for complex data analysis                                                     |
| CO4                                               | Design and optimize neural network models using forward and backpropagation, activation functions, regularization, and hyperparameter tuning techniques for real-world applications. |
| CO5                                               | Analyze and optimize machine learning models using ensemble methods, reinforcement learning techniques, and parameter tuning approaches for real-world applications.                 |

| Text Books:      |                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1                | Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", 2nd Edition, shroff publisher, 2022.                 |
| 2                | Ethem Alpaydin, "Introduction to Machine Learning", Fourth Edition, MIT Press, 2020.                                                       |
| Reference Books: |                                                                                                                                            |
|                  |                                                                                                                                            |
| 1                | Sebastian Raschka and Vahid Mirjalili, <i>Python Machine Learning</i> , 3rd Edition, Packt Publishing, 2022.                               |
| 2                | Jason Bell, "Machine learning - Hands on for Developers and Technical Professionals", First Edition, Wiley, 2014.                          |
| 3                | Stephen Marsland, "Machine Learning: An Algorithmic Perspective", Second Edition, CRC Press, 2014.                                         |
| 4                | Peter Flach "Machine Learning: The Art and Science of Algorithms that Make Sense of Data" First Edition, Cambridge University Press, 2012. |

| Online Resources: |                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                 | <a href="https://online.stanford.edu/courses/soe-ymls-machine-learning-specialization">https://online.stanford.edu/courses/soe-ymls-machine-learning-specialization</a> |
| 2                 | <a href="https://developers.google.com/machine-learning/crash-course">https://developers.google.com/machine-learning/crash-course</a>                                   |
| 3                 | <a href="https://www.harvardonline.harvard.edu/course/data-science-machine-learning">https://www.harvardonline.harvard.edu/course/data-science-machine-learning</a>     |
| 4                 | <a href="https://onlinecourses.nptel.ac.in/noc25_cs91/preview">https://onlinecourses.nptel.ac.in/noc25_cs91/preview</a>                                                 |
| 5                 | <a href="https://github.com/compops/pmh-tutorial">https://github.com/compops/pmh-tutorial</a>                                                                           |

| CO/PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 | PSO3 |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|
| CO1   | 3   | 3   | -   | -   | 3   | -   | -   | 2   | 2   | -    | 3    | 3    | 2    | -    |
| CO2   | 3   | 3   | 2   | -   | 3   | -   | 2   | 2   | 2   | -    | 3    | 3    | 3    | 2    |
| CO3   | 3   | 3   | 2   | -   | 3   | -   | 2   | 2   | 2   | -    | 3    | 3    | 3    | 2    |
| CO4   | 3   | 3   | 3   | -   | 3   | -   | 2   | 2   | 2   | -    | 3    | 3    | 3    | 3    |
| CO5   | 3   | 3   | 3   | 2   | 3   | -   | 2   | 2   | 2   | -    | 3    | 3    | 3    | 3    |

Correlation levels 1, 2 and 3 are as defined below: 1 - Slight 2 Moderate 3 - Substantial (High)

HoD

Dean (Academics)

Principal